Panel Data Analysis

School of Management and Economics
The Chinese University of Hong Kong,
Shenzhen

Notations for Panel Data

A double subscript is used to distinguish entities (individuals, states, etc.) and time periods (months, years, etc.):

- i = entity, n = the total number of entities, so $i = 1, \ldots, n$
- t = time period, T = the total number of time periods, so $t = 1, \ldots, T$

Data: Suppose we have k regressors. The data are:

$$(X_{1it}, X_{2it}, \ldots, X_{kit}, Y_{it}), i = 1, \ldots, n, t = 1, \ldots, T .$$

- Balanced panel: no missing observations
- Unbalanced panel: some entities (individuals, states, etc.) are not observed for some time periods (months, years, etc.)

Why Are Panel Data Useful?

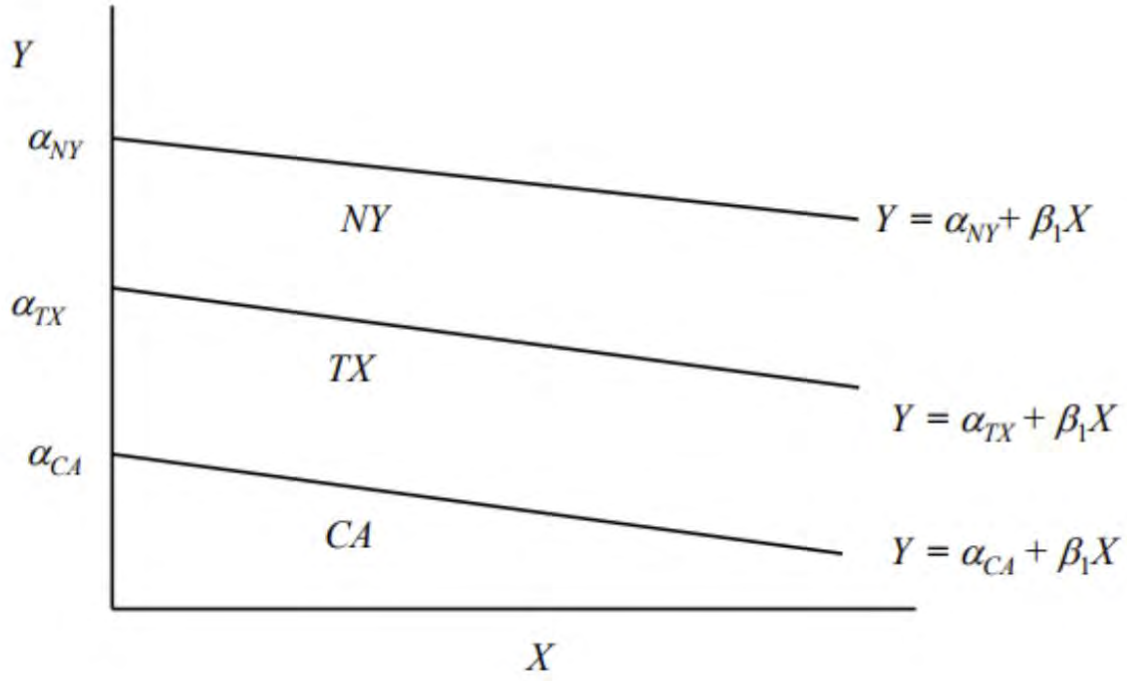
With panel data we can control for factors that:

- Are unobserved or unmeasured – and therefore cannot be included in the regression using multiple regression
- Could cause omitted variable bias if they are omitted
- Vary across entities but do not vary over time

Here's the key idea:

If an omitted variable does not change over time, then any changes in Y over time cannot be caused by the omitted variable – the omitted variable can be removed by some methods.

Fixed Effects Form



Both State and Time Fixed Effects

Estimation:

1. "$n-1$ and $T-1$ binary regressor" OLS regression
   - First create the binary variables $D2_i, \ldots, Dn_i, B2_t, \ldots, BT_t$
   - Then estimate the model by OLS
   - Inference (hypothesis tests, confidence intervals) is as usual (using heteroskedasticity-robust standard errors)
2. "Entity and time-demeaned" OLS regression
   - Deviate Y_{it}, X_{it} from entity averages
   - Deviate the deviations in step 1 from time averages
   - Estimate by OLS using "time- and entity-demeaned" data
   - This can be easily done in STATA
3. "Changes" OLS regression

Treatment effect

The term ‘treatment effect’ refers to the causal effect of a binary (0–1) variable on an outcome variable of scientific or policy interest.

Economics examples include the effects of government programmes and policies, such as those that subsidize training for disadvantaged workers, and the effects of individual choices like college attendance.

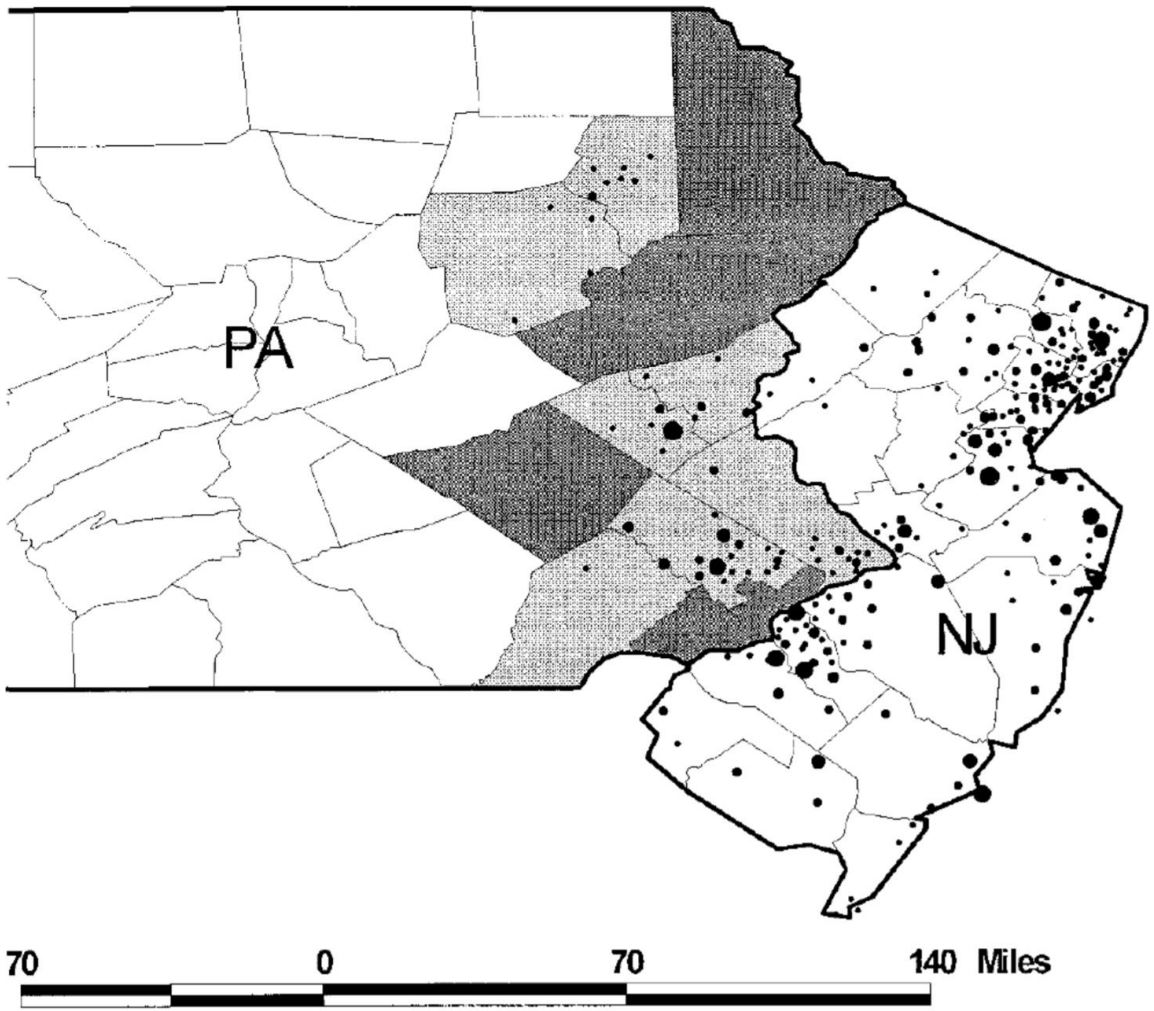
Selection on Unobservables

- Often there are reasons to believe that treated and untreated units differ in unobservable characteristics that are associated with potential outcomes even after controlling for differences in observed characteristics.
- In such cases, treated and untreated units are not directly comparable. What can we do then?

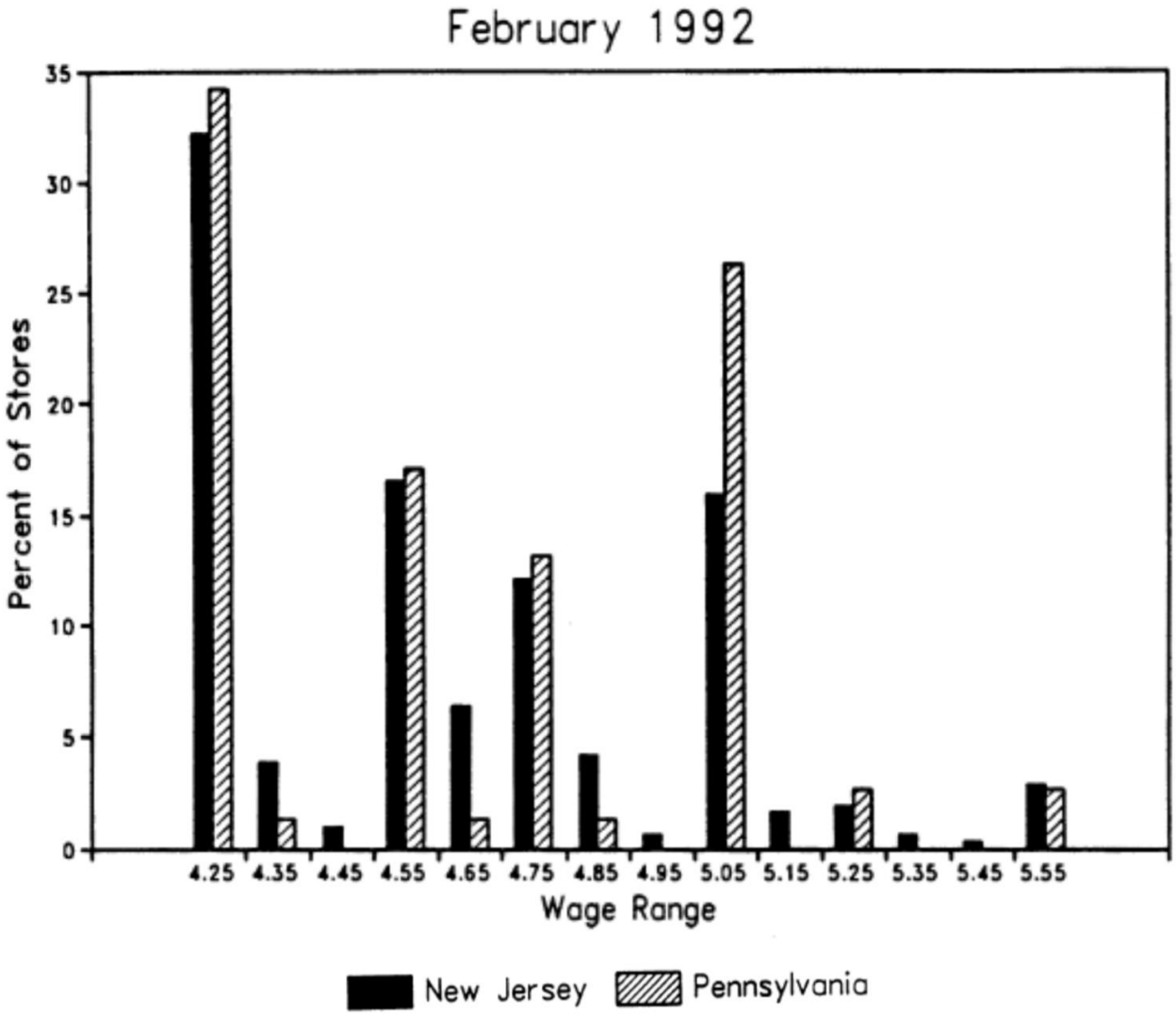
Example: Minimum wage laws and employment

- Do higher minimum wages decrease low-wage employment?
- Card and Krueger (1994) consider impact of New Jersey's 1992 minimum wage increase from \$4.25 to \$5.05 per hour
- Compare employment in 410 fast-food restaurants in New Jersey and eastern Pennsylvania before and after the rise
- Survey data on wages and employment from two waves:
  - Wave 1: March 1992, one month before the minimum wage increase
  - Wave 2: December 1992, eight months after increase

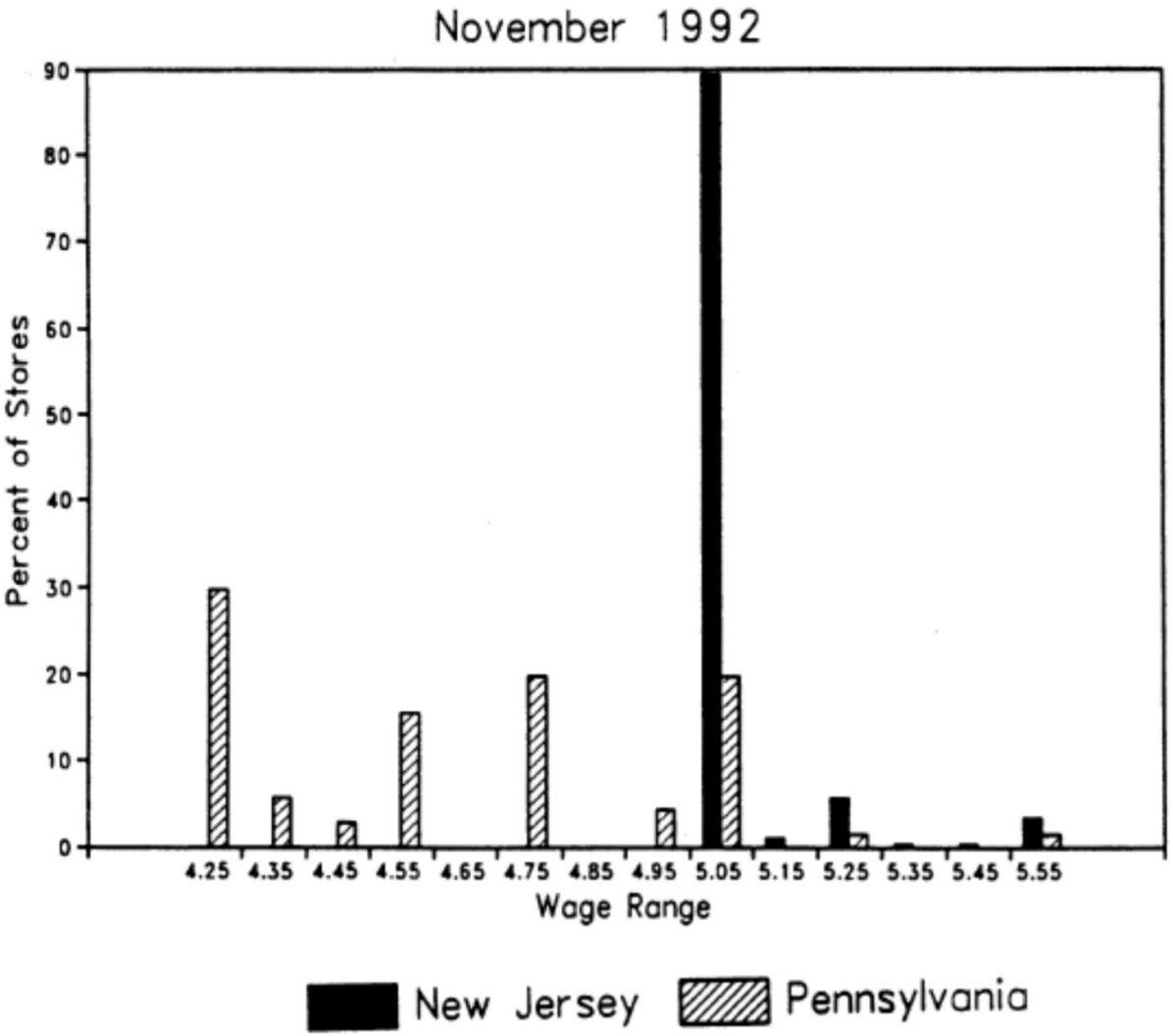
Locations of Restaurants (Card and Krueger 2000)



Wages Before Rise in Minimum Wage



Wages After Rise in Minimum Wage



Example

- Question: What is the effect of a decline in expected bankruptcy costs on corporate debt usage?
  - Tax-bankruptcy cost theories of capital structure predict that debt usage and expected bankruptcy costs are inversely related
- Ideal but Impossible Experiment:
  - Take a set of firms, reduce bankruptcy costs (e.g., streamline bankruptcy procedures) and measure debt usage
  - "Rewind the clock," take the same set of firms and measure their debt usage.
  - Compare debt usage across two scenarios
- Desirable but Infeasible Experiment
  - Take a set of firms and randomly select some fraction of firms to be subject to the new bankruptcy procedures.
  - Compare debt usage across the two sets of firms

What was Good about these Experiments?

- The key to program evaluation is estimating the counterfactual: What would have happened had the treated not be treated?
- Therefore, quality of our evaluation is tied to how well we can estimate the counterfactual
  - » The Ideal but Impossible Experiment actually provides the counterfactual by "rewinding the clock."
  - » The Desirable but Infeasible Experiment provides a good estimate of the counterfactual by the random assignment
    - There are no systematic differences between the treated and untreated groups that are related to the outcome of interest
    - Without random assignment (i.e., self-selection or manipulation) then we can't be sure if differences (or similarities) between the treated and untreated are due to the program or some other difference between the two groups.

The Natural Experiment

- At the end of 1991, Delaware passes a law that significantly streamlines bankruptcy proceedings to make litigation less costly and time-consuming.
  - » Must assume that this is a *random* (or outcome unrelated) event.
    - The event should not be a response to pre-existing differences between the treatment and control group (e.g., Delaware firms are much more likely to enter financial distress)
      - E.g., "Ashenfelter Dip" (Ashenfelter and Card (1985))
  - » Must understand what *caused* the event to occur
- This law change offers a potentially useful setting with which to test our hypothesized relation between bankruptcy costs and capital structure
  - » How do we empirically test the relation?

Cross-Sectional Difference After Treatment

- Let's compare the average leverage of firms registered in Delaware to that of firms registered elsewhere
  - This can be accomplished via a cross-sectional regression

$$y_i = \beta_0 + \beta_1 I\left(treat_i\right) + \varepsilon_i$$

  - where y_i = Leverage for firm i in 1992, and $I(treat_i) = 1$ if firm is registered in Delaware
- Assuming $E(\varepsilon_i \mid I(treat_i)) = 0$:

$$E\left(y_i \mid I\left(treat_i\right) = 0\right) = \beta_0$$

$$E\left(y_i \mid I\left(treat_i\right) = 1\right) = \beta_0 + \beta_1$$

$$\Rightarrow E\left(y_i \mid I\left(treat_i\right) = 1\right) - E\left(y_i \mid I\left(treat_i\right) = 0\right) = \beta_1$$

- Our estimate is just the difference in average leverage in 1992 for the treatment group (Delaware firms) and control group (non-Delaware firms)

Cross-Sectional Difference After Treatment

- As usual, the concern lies with our assumption of $E(\varepsilon_i \mid I(treat_i)) = 0$
- What could threaten this assumption and, consequentially, the internal validity of our estimate?
  - What if firms in Delaware are in more capital intensive industries relative to firms elsewhere?
    - Problem is that firms with more physical capital tend to be more levered so our assumption is violated because capital intensity is sitting in ε and it's correlated with treatment status
    - In other words, even if the law was never passed, we would expect firms in Delaware to have higher leverage then other firms because of differences in capital intensity

Cross-Sectional Difference After Treatment

- This is just an omitted variables bias where the true relation is:

$$y_i = \beta_0 + \beta_1 I(treat_i) + \beta_2 CI_i + \eta_i$$

but we estimate

$$y_i = \beta_0 + \beta_1 I(treat_i) + \varepsilon_i$$

- Implies the OLS estimator of β_1 is:

$$\hat{\beta}_1 = \frac{Cov(I(treat), y)}{Var(I(treat))} = \frac{Cov(I(treat), \beta_0 + \beta_1 I(treat) + \beta_2 CI + \eta)}{Var(I(treat))}$$

$$= \beta_1 + \beta_2 \frac{Cov(I(treat), CI)}{Var(I(treat))} = \beta_1 + \beta_2 \gamma$$

where γ is the slope coefficient from a regression of capital intensity *CI* on *I*(*treat*)

- $\gamma > 0, \beta_2 > 0 \rightarrow$ OLS estimate biased up in favor of a treatment effect…not good.

Cross-Sectional Difference After Treatment

- One Solution:
  - Insert a proxy for capital intensity (e.g., net plant, property, and equipment divided by assets) in the regression. But,
    - this is just a proxy and just one variable
  - There is still some heterogeneity between treatment and control groups that is sitting in the error term and is correlated with our treatment indicator
    - A question of magnitude but we can't estimate this since error unobservable

Cross-Sectional Difference After Treatment

- What's common across these examples of threats to internal validity is that they correspond to unmeasured differences between the two groups
  - » Imagine a latent fixed effect across the two groups
    - This results from non-random assignment to treatment and control
  - » This will confound any comparison between the two groups

Time-Series Difference Within the Treatment Group

- Let's compare the average leverage of firms registered in Delaware in 1991 to that in 1992
  - This can be accomplished via a two-period panel regression using *only* Delaware firms

$$y_{it} = \beta_0 + \beta_1 I\left(Post_{it}\right) + \varepsilon_{it}$$

where $I(Post) = 1$ if year = 1992 and 0 if year = 1991.

  - or, after differencing, a cross-sectional regression using *only* Delaware firms

$$\Delta y_i = \beta_1 + \Delta \varepsilon_i$$

Assuming $E(\varepsilon_{it} \mid I(Post_{it})) = 0$

$$E\left(y_{it} \mid I\left(Post_{it}\right) = 0\right) = \beta_0$$

$$E\left(y_{it} \mid I\left(Post_{it}\right) = 1\right) = \beta_0 + \beta_1$$

$$\Rightarrow E\left(y_{it} \mid I\left(Post_{it}\right) = 1\right) - E\left(y_{it} \mid I\left(Post_{it}\right) = 0\right) = \beta_1$$

Time-Series Difference Within the Treatment Group

- The primary concern here is other factors affecting leverage *over time*.
  - » E.g., Increase in the supply of credit due to financial innovation
  - » Leverage would likely have increased for firms even without the passage of the law
  - » Bias is positive → in favor of supporting a treatment effect
- This is just another form of omitted variables bias, as before
  - » We could insert control variables but difficult to measure all perfectly
- Also, bias can work in both directions
  - » E.g., 1992 may be a period of tight credit, which leads to a decline in debt usage and offsetting effect

Difference in Differences: Setup

Definition

Two groups:

- $D = 1$ Treated units
- $D = 0$ Control units

Two periods:

- $T = 0$ Pre-Treatment period
- $T = 1$ Post-Treatment period

Potential outcomes $Y_d(t)$:

- $Y_{1i}(t)$ potential outcome unit i attains in period t when treated between t and $t-1$
- $Y_{0i}(t)$ potential outcome unit i attains in period t with control between t and $t-1$

Difference in Differences: Setup

Definition

Causal effect for unit i at time t is

- $\tau_{it} = Y_{1i}(t) - Y_{0i}(t)$

Observed outcomes $Y_i(t)$ are realized as

- $Y_i(t) = Y_{0i}(t) \cdot (1 - D_i(t)) + Y_{1i}(t) \cdot D_i(t)$

Fundamental problem of causal inference:

- If D only occurs when $t = 1$, then for the treatment group $(D_i = 1)$ we have: $Y_i(1) = Y_{0i}(1) \cdot (1 - D_i(1)) + Y_{1i}(1) \cdot D_i(1)$
- $= Y_{1i}(1)$ (i.e. we don't get to see $Y_{0i}(1)$ for the treated)

Estimand (ATT)

Focus on estimating the average effect of the treatment on the treated: $\tau_{ATT} = E[Y_{1i}(1) - Y_{0i}(1) | D = 1]$

Two Groups and Two Periods

Estimand (ATT)

$\tau_{ATT} = E[Y_{1i}(1) - Y_{0i}(1)|D = 1]$

	Post-Period (T=1)	**Pre-Period (T=0)**
Treated D_i=1	$E[Y_{1i}(1)\|D_i = 1]$	$E[Y_{0i}(0)\|D_i = 1]$
Control D_i=0	$E[Y_{0i}(1)\|D_i = 0]$	$E[Y_{0i}(0)\|D_i = 0]$

Problem

Missing potential outcome: $E[Y_{0i}(1)|D = 1]$*, ie. what is the average post-period outcome for the treated in the absence of the treatment?*

Two Groups and Two Periods

Estimand (ATT)

$$\tau_{ATT} = E[Y_{1i}(1) - Y_{0i}(1)|D = 1]$$

	Post-Period (T=1)	**Pre-Period (T=0)**
Treated D_i=1	$E[Y_{1i}(1)\|D_i = 1]$	$E[Y_{0i}(0)\|D_i = 1]$
Control D_i=0	$E[Y_{0i}(1)\|D_i = 0]$	$E[Y_{0i}(0)\|D_i = 0]$

Control Strategy: Before vs. After

- Use: $E[Y_i(1)|D_i = 1] - E[Y_i(0)|D_i = 1]$
- Assumes: $E[Y_{0i}(1)|D_i = 1] = E[Y_{0i}(0)|D_i = 1]$

Two Groups and Two Periods

Estimand (ATT)

$\tau_{ATT} = E[Y_{1i}(1) - Y_{0i}(1)|D = 1]$

	Post-Period (T=1)	**Pre-Period (T=0)**
Treated D_i=1	$E[Y_{1i}(1)\|D_i = 1]$	$E[Y_{0i}(0)\|D_i = 1]$
Control D_i=0	$E[Y_{0i}(1)\|D_i = 0]$	$E[Y_{0i}(0)\|D_i = 0]$

Control Strategy: Treated-Control Comparison in Post-Period

- Use: $E[Y_i(1)|D_i = 1] - E[Y_i(1)|D_i = 0]$
- Assumes: $E[Y_{0i}(1)|D_i = 1] = E[Y_{0i}(1)|D_i = 0]$

Two Groups and Two Periods

Estimand (ATT)

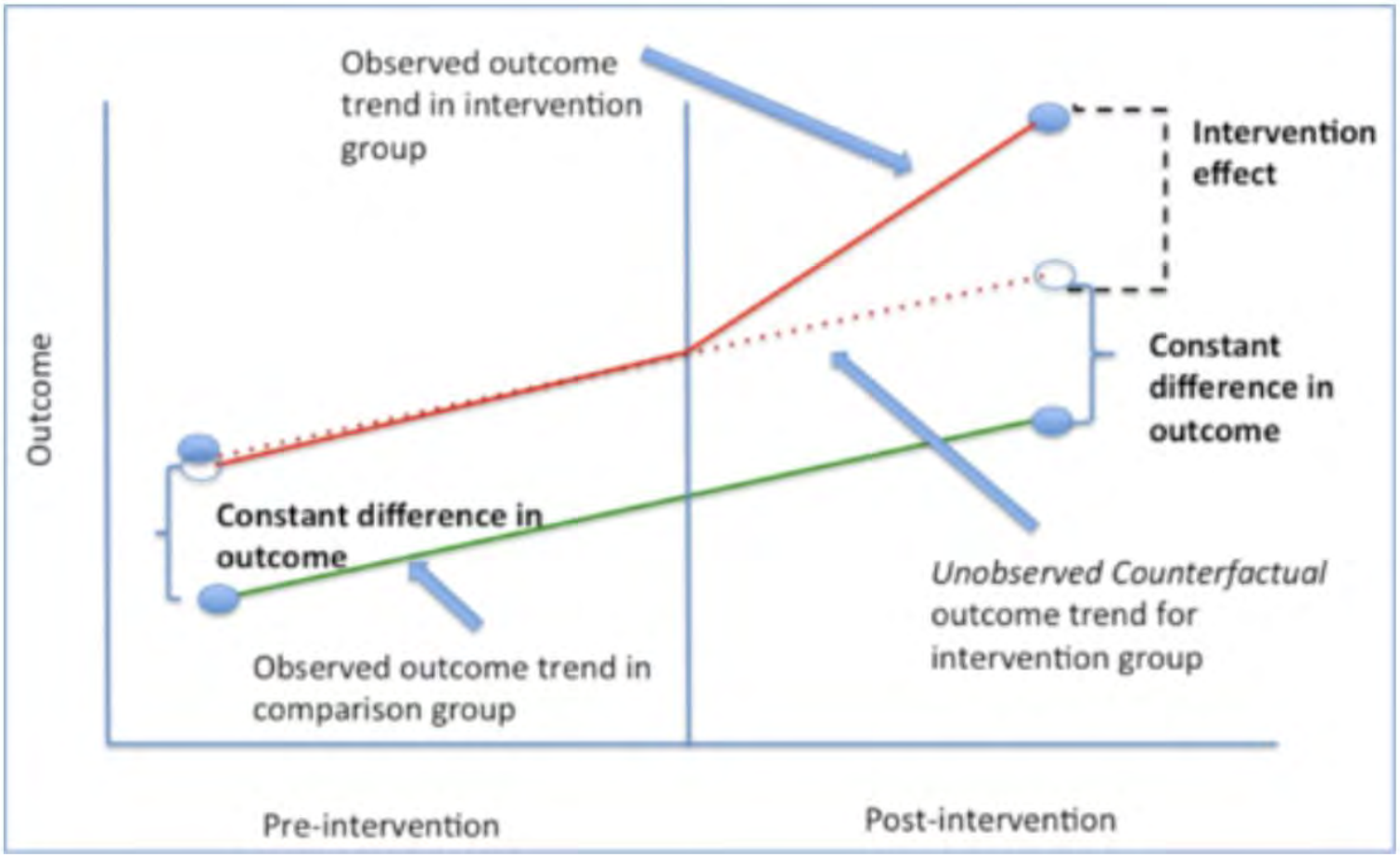
$\tau_{ATT} = E[Y_{1i}(1) - Y_{0i}(1)|D = 1]$

	Post-Period (T=1)	**Pre-Period (T=0)**
Treated D_i=1	$E[Y_{1i}(1)\|D_i = 1]$	$E[Y_{0i}(0)\|D_i = 1]$
Control D_i=0	$E[Y_{0i}(1)\|D_i = 0]$	$E[Y_{0i}(0)\|D_i = 0]$

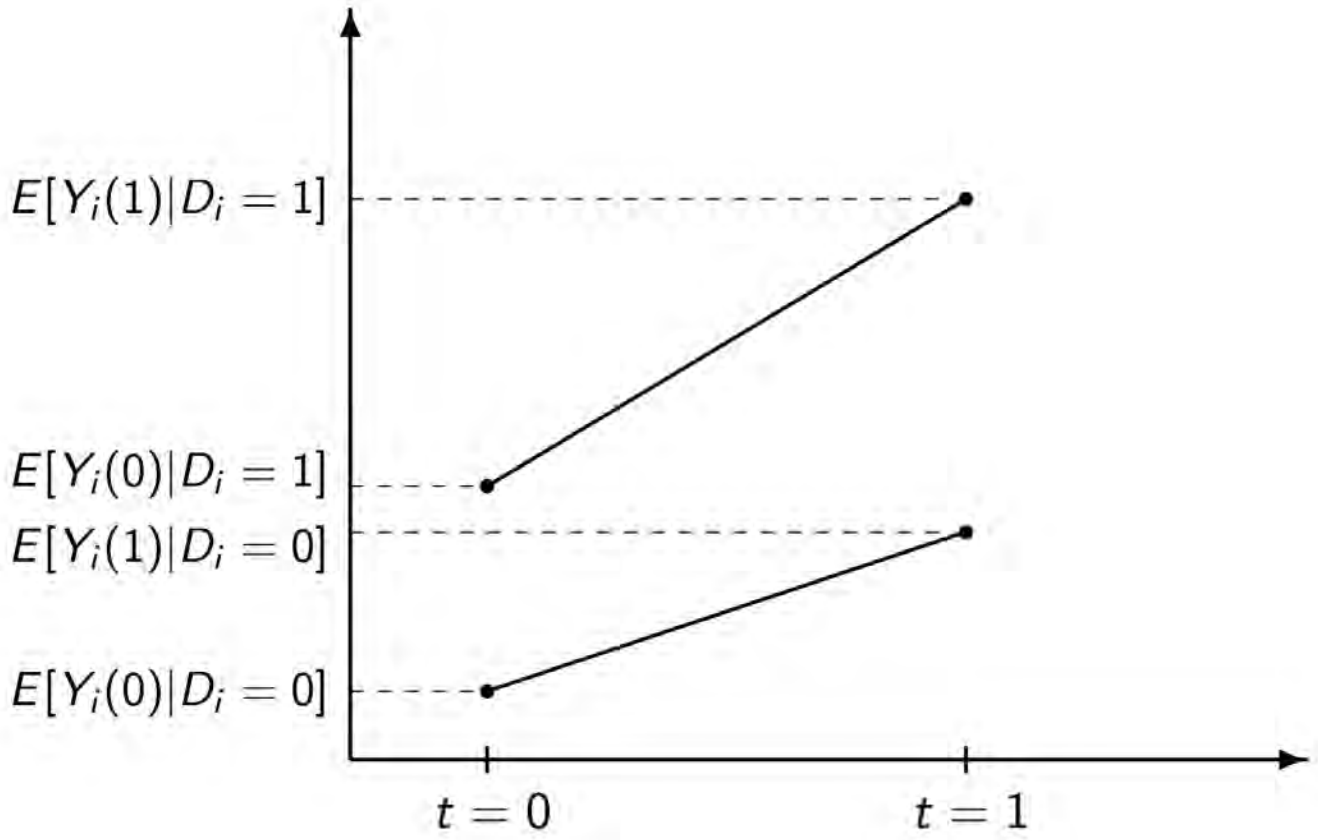
Control Strategy: Difference-in-Differences (DD)

- Use:
  $\Big\{E[Y_i(1)|D_i = 1] - E[Y_i(1)|D_i = 0]\Big\} -$
  $\Big\{E[Y_i(0)|D_i = 1] - E[Y_i(0)|D_i = 0]\Big\}$
- Assumes: $E[Y_{0i}(1) - Y_{0i}(0)|D_i = 1] = E[Y_{0i}(1) - Y_{0i}(0)|D_i = 0]$

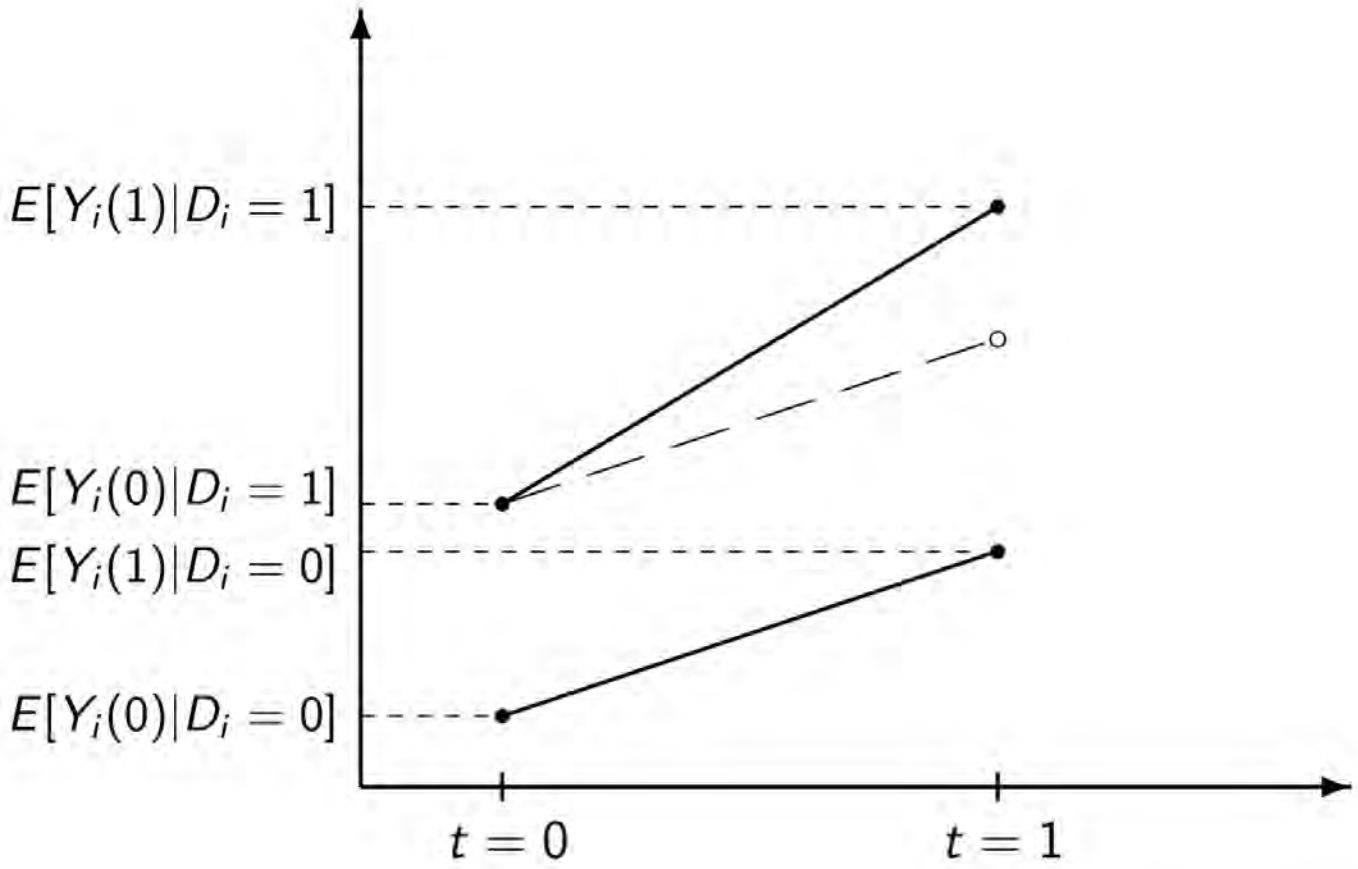
Graphical Representation: Difference-in-Differences



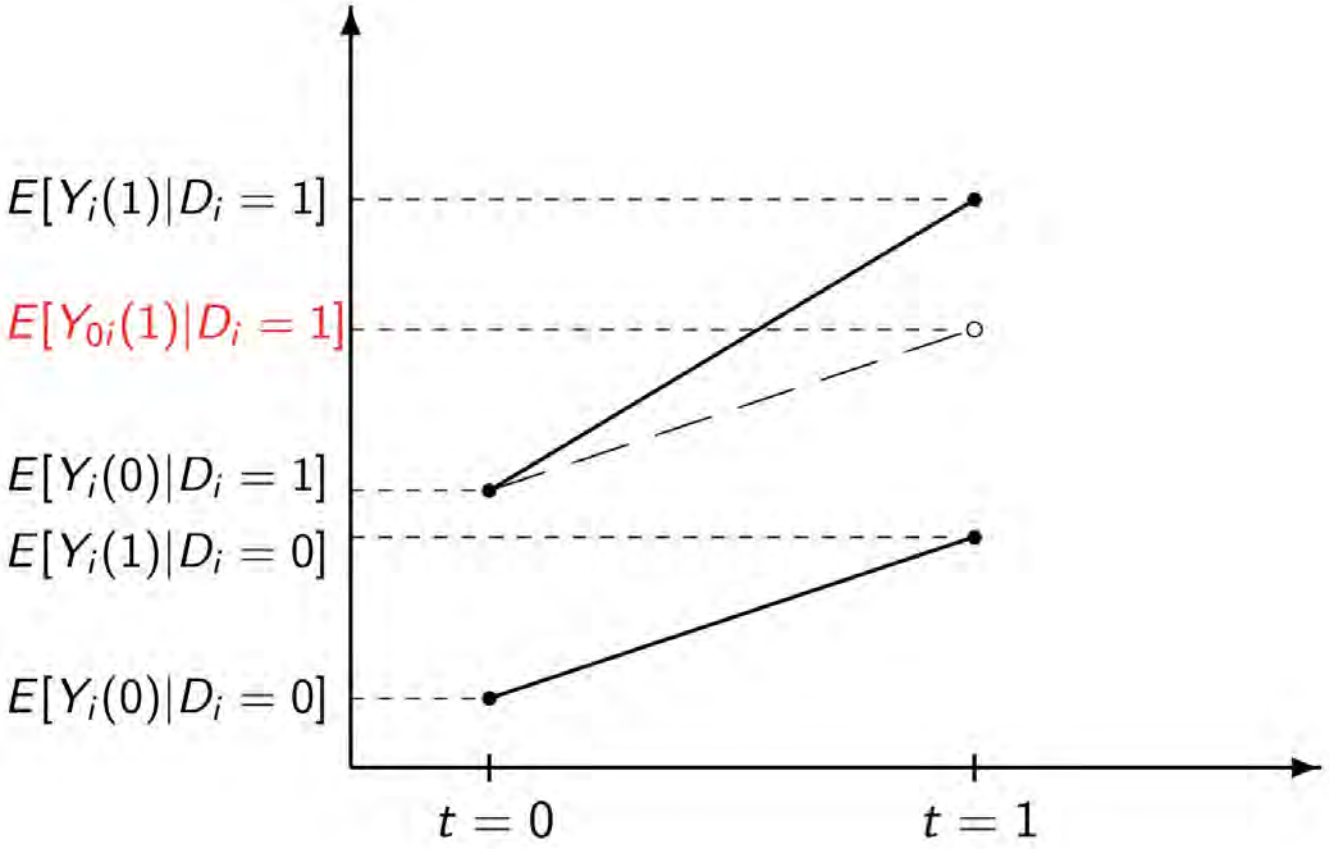
Graphical Representation: Difference-in-Differences



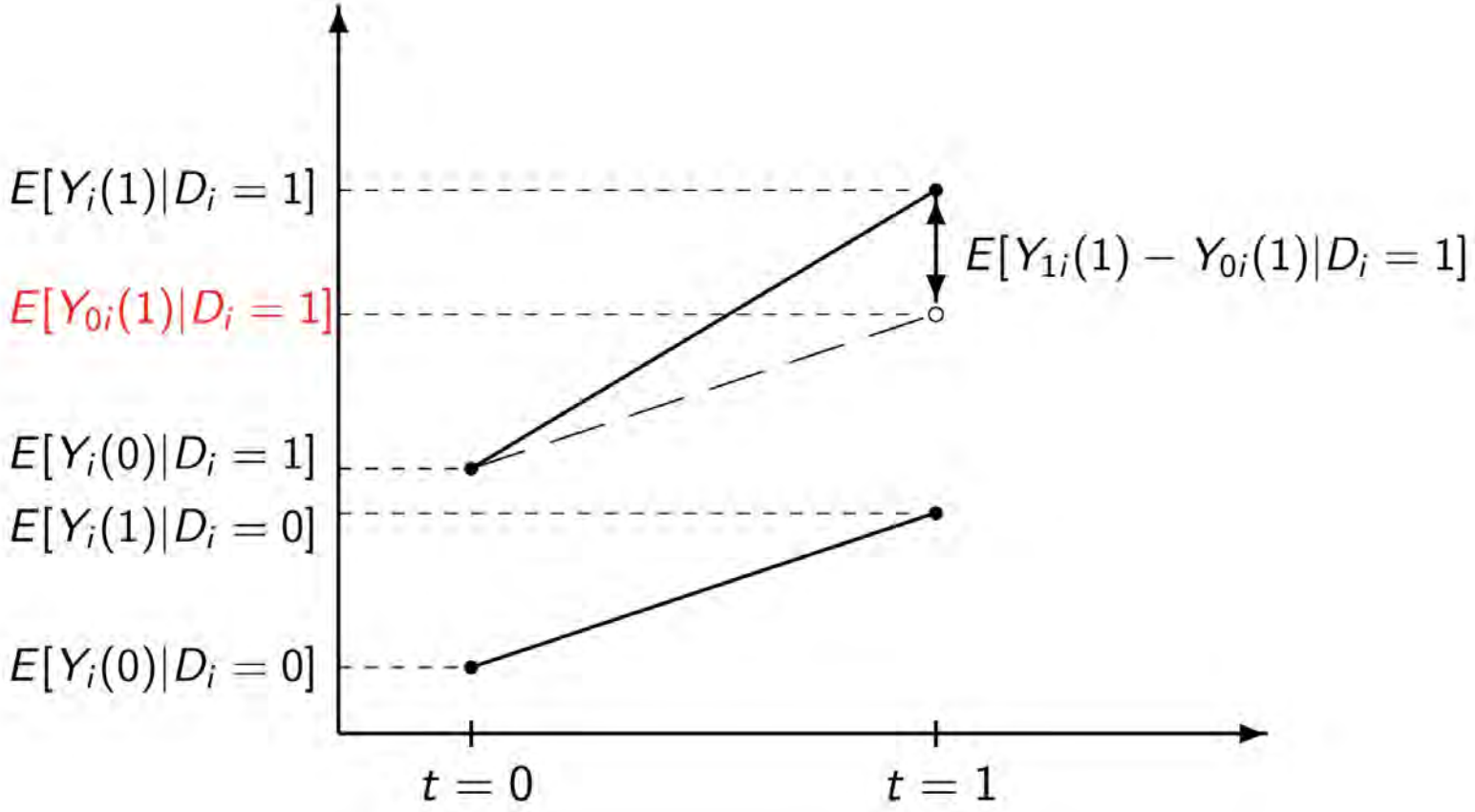
Graphical Representation: Difference-in-Differences



Graphical Representation: Difference-in-Differences



Graphical Representation: Difference-in-Differences



DD Estimator

- It would be nice to combine the positive features of each single difference estimator
  - Cross-sectional estimator avoided omitted common trends
  - Time-series estimator avoided omitted cross-sectional differences
- The DD estimator does precisely that

$$y_{it} = \beta_0 + \beta_1 I(treat_{it}) + \beta_2 I(post_{it}) + \beta_3 I(treat_{it}) \times I(post_{it}) + \varepsilon_{it}$$

  - We need a full panel of firms consisting of Delaware ($I(treat) = 1$) and Non-Delaware ($I(treat) = 0$) registered firms observed before ($I(post) = 0$) and after ($I(post) = 1$) the passage of the law
  - β_3 is the parameter of interest (i.e., the DD estimator)

DD Estimator

- Easier to see what the parameters estimate by arranging things in a 2 x 2 matrix:

	Post	*Pre*	*Difference*
Treatment	$\beta_0 + \beta_1 + \beta_2 + \beta_3$	$\beta_0 + \beta_1$	$\beta_2 + \beta_3$
Control	$\beta_0 + \beta_2$	β_0	β_2
Difference	$\beta_1 + \beta_3$	β_1	β_3

- Note:
  - Implies alternative route to DD estimator: subtract difference in average leverage for treatment and control in 1991 from the difference in average leverage for treatment and control in 1992
  - β_1 is the average leverage of treatment firms in the pre-treatment era *relative to* (i.e. minus) the average leverage of control firms in the pre-treatment era
  - β_2 is the average leverage of control firms in the post-treatment era *relative to* (i.e. minus) the average leverage of control firms in the pre-treatment era

Re-Visiting the Single Difference Estimators

- Cross-sectional difference:

$$E\left(y \mid I\left(treat\right)=1, I\left(post\right)=1\right)-E\left(y \mid I\left(treat\right)=0, I\left(post\right)=1\right)=\beta_1+\beta_3$$

  - Unbiased if β_1 is equal to zero → no permanent difference between the treatment and control groups

- Time-series difference:

$$E\left(y \mid I\left(treat\right)=1, I\left(post\right)=1\right)-E\left(y \mid I\left(treat\right)=1, I\left(post\right)=0\right)=\beta_2+\beta_3$$

  - Unbiased if β_2 is equal to zero → no common trend over the pre- and post-treatment eras

- DD estimator avoids these two threats by differencing away any permanent differences between the groups *and* any common trend affecting both groups

Key Assumption Behind DD

- In the absence of treatment, the average change in the response variable would have been the same for both the treatment and control groups.
  - a.k.a. the **parallel trends** assumption since it requires that the trend in the outcome variable for both treatment and control groups during the pre-treatment era are similar.
- What this assumptions does *not* mean
  - It does *not* mean that the there is *no* trend in the outcome variable during the pre-treatment era (just the same trend across groups)
  - It does *not* require that the *level* of the outcome variable for the two groups be same in the pre-treatment era
    - This can create other issues concerning function form…more on this later